Course Outline for CSE-346

Part A

1. Course Code: CSE-346

2. Course Title: Artificial Intelligence with Python (Sessional)

3. Course Type: Core Course

4.Level/ Term: Level: 3 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Dr. Mohammed Moshiul Hoque, Professor, Dept. of CSE, CUET
Annesha Das, Lecturer, Dept. of CSE, CUET
Ashim Dey, Lecturer, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 1.5 (3 hours of lab work per week)

9. Contact Hours: 3 hours of lab work per week

10. Total Marks: 150

11. Rational of the Course:

This course will cover the basic of python programming, syntax and semantics of Python, implementation and analysis of AI related algorithms. This course is important to understand the fundamental of python programming and how python can be used to solve real world complex problem. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Sessional based on the following topics with emphasis on Python Programming

Introduction to old and new AI techniques; Intelligent agents; Search techniques in AI; Constraint satisfaction problems; Game playing; Knowledge representation and reasoning; Propositional and first order logic, inference in first order logic; Planning; Uncertainty; Probabilistic reasoning; Learning in symbolic and non-symbolic representation; Expert systems and knowledge engineering; Natural language processing; Computer vision; Robotics;

12. Course Objectives:

(a) Familiarization with syntax and semantics of Python.

(b) Developing the knowledge of real world complex problem solving with Python.

(c) Acquiring skills of building AI based projects using Python.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	Effectively use the resources of Python to solve a problem	C3, A1
CLO2	Apply different search strategies to achieve a certain goal from initial situation in case of building rational agents	C3
CLO3	Propose and design a solution of a real world problem	C5, A3,A5,P7

b) Mapping of CLOs with PLOs

No.	CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1	X											
2	CLO2			X									
3	CLO3			X		X		X					

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to Python and discussion on importance of Python in Artificial Intelligence & IDE selection	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1

Week - 02	Familiarization with syntax and semantics of Python.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 03	Familiarization with Python packages and modules.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 04	Study of OOP and exception handling using Python	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO1
Week - 05	Project idea discussion	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-3
Week - 06	Implementation of Uninformed search: BFS & DFS using Python	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 07	Implementation of Uninformed search: IDS search using Python	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation	Lab Performance Report	• CLO-2

		• Group Discussion		
Week - 08	Implementation of Informed search: Greedy best first search using Python.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 09	Implementation of Informed search: A* search algorithm using Python.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 10	Implementation of 8-puzzle problem using Python	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 11	Implementation of minimax algorithm using python	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 12	Lab Test and Project Presentation			
Week - 13	Viva-voce Quiz			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%

b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- Stuart Russel and Peter Norvig-- Artificial Intelligence A Modern Approach by Prentice Hall (2009, 3rd Edition)

2) Supplementary Readings

- Elaine Rich and Kevin Knight -- Artificial Intelligence by Tata McGraw Hill (2017, 3rd Edition)
- Dan W. Patterson -- Introduction to Artificial Intelligence & Expert Systems by Prentice Hall (1990)

3) Others

- Laboratory Materials